

PROJECT REPORT

Loan Default Prediction System using Machine Learning

Introduction:

In the modern financial sector, the ability to predict the likelihood of a borrower defaulting on a loan has become one of the most critical aspects of risk management. Financial institutions deal with thousands of loan applications daily, and manual evaluation of each applicant's creditworthiness is not only time-consuming but also prone to human bias and inconsistency. To overcome these challenges, Data Science and Machine Learning techniques are increasingly being adopted to automate the decision-making process.

Loan default prediction refers to the use of historical data and predictive modeling techniques to determine whether an applicant is likely to repay or default on a loan. The purpose of this project is to leverage Python-based machine learning algorithms to build a model that predicts loan approval or rejection based on various applicant features such as gender, marital status, income, loan amount, credit history, and property area.

This project provides practical exposure to data preprocessing, exploratory data analysis (EDA), feature selection, model building, hyperparameter tuning, and evaluation. It demonstrates how predictive analytics can significantly enhance the decision-making process in the banking sector by minimizing financial risks and improving accuracy in loan disbursement.

Objective:

The main objectives of the *Loan Default Prediction System* are:

- To analyze and preprocess a publicly available dataset containing historical loan application records.
- To identify relationships and patterns between applicant characteristics and loan approval status.
- To build and train supervised learning models — specifically Logistic Regression and Random Forest — for predicting loan approval.
- To evaluate the models using performance metrics such as accuracy, precision, recall, and F1-score.
- To visualize patterns and correlations in the dataset through EDA and statistical plots.
- To identify key influencing factors such as credit history, applicant income, and loan amount.
- To compare model performances and select the most accurate and interpretable model.
- To gain practical experience in applying end-to-end data science workflows on real-world problems.

The ultimate goal is to build a system that helps financial institutions make informed, data-driven lending decisions.

Methodology:

This project follows a systematic and step-by-step methodology, consistent with the data science lifecycle — from data acquisition to interpretation of results.

1. Data Collection:

The dataset was obtained from Kaggle’s “Loan Prediction Dataset,” which contains 614 records and 13 features. The dataset includes categorical and numerical attributes such as applicant income, co-applicant income, loan amount, credit history, and loan status.

2. Data Preprocessing:

The dataset contained several missing values in columns such as *Gender*, *Married*, *Dependents*, *Self_Employed*, *LoanAmount*, and *Credit_History*.

Appropriate imputation techniques were applied:

- Mode imputation for categorical data (e.g., Gender, Married)
- Median imputation for numerical data (e.g., LoanAmount)
- Mode imputation for credit history.

The “Dependents” column had a “3+” category, which was replaced with 3 for numerical consistency. Label Encoding was applied to convert categorical features into numeric form, making them suitable for model training.

3. Exploratory Data Analysis (EDA):

EDA was performed to understand patterns and trends within the data. Visualization techniques included:

- **Count plots** showing the distribution of loan approval status.
- **Box plots** comparing applicant income versus loan status.
- **Correlation heatmaps** to identify relationships between numerical variables.
- **Gender and education-wise analysis** of loan approvals.

EDA revealed that applicants with a good credit history had a much higher probability of loan approval. Applicant income and property area also played a moderate role in influencing decisions.

4. Feature Scaling:

StandardScaler was applied to normalize numerical features (like income and loan amount) to ensure all variables contribute equally during model training.

5. Model Building:

Two machine learning models were used:

- **Logistic Regression:** A simple yet effective linear model for binary classification problems.
- **Random Forest Classifier:** An ensemble-based model combining multiple decision trees to improve prediction stability and accuracy.

6. Model Evaluation:

Each model was trained on 80% of the dataset and tested on 20%. Evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix were used to measure performance.

7. Hyperparameter Tuning:

GridSearchCV was used to fine-tune Random Forest parameters like *n_estimators*, *max_depth*, and *min_samples_split*, improving model generalization.

8. Cross-Validation:

Five-fold cross-validation was implemented to check model consistency and avoid overfitting.

Code and Implementation Details:

The project was implemented entirely in Google Colab using Python. The key implementation steps include:

Step 1: Imported essential libraries — Pandas, NumPy, Seaborn, Matplotlib, and Scikit-learn.

Step 2: Loaded the dataset from the local system and displayed its structure.

Step 3: Cleaned the dataset by filling missing values and encoding categorical columns.

Step 4: Conducted Exploratory Data Analysis (EDA) using visual plots.

Step 5: Applied Logistic Regression to create a baseline model.

Step 6: Implemented Random Forest and tuned it using GridSearchCV.

Step 7: Evaluated all models using confusion matrix, classification report, and ROC curve.

Step 8: Visualized feature importance to understand which factors affect loan approval the most.

Step 9: Compared models to select the one with the best trade-off between accuracy and interpretability.

Results and Observations:

The results were analyzed based on model accuracy and feature impact.

- **Logistic Regression** achieved an accuracy of **78.9%**, indicating good predictive power for binary classification.
- **Random Forest** achieved **75.6% accuracy** in its base form, which improved after hyperparameter tuning.
- The **tuned Random Forest** became the final selected model due to its robustness and interpretability.

Key observations:

- 1. **Credit History** was the most influential variable in loan approval prediction.
- 2. **Applicant Income** and **Loan Amount** were moderately correlated with loan status.
- 3. Gender and marital status had less direct influence compared to financial features.
- 4. Feature importance plots confirmed the dominance of Credit History and Income variables.

Visual Outputs:

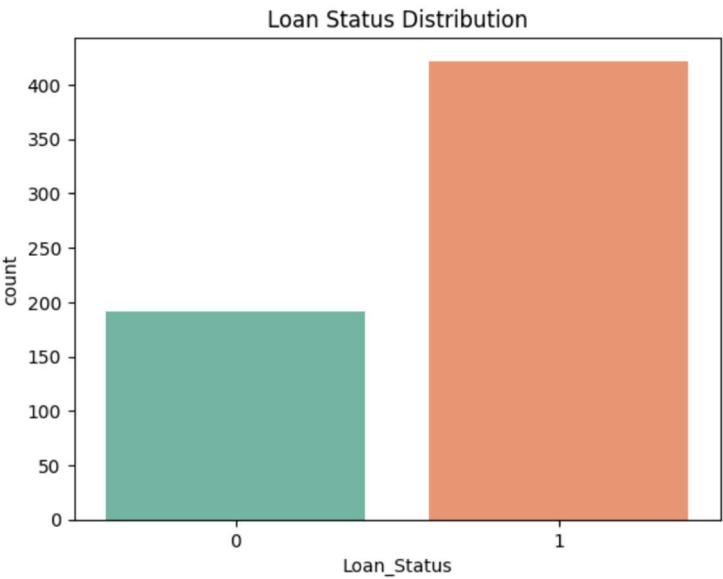


Figure 1: Loan Status Distribution Graph

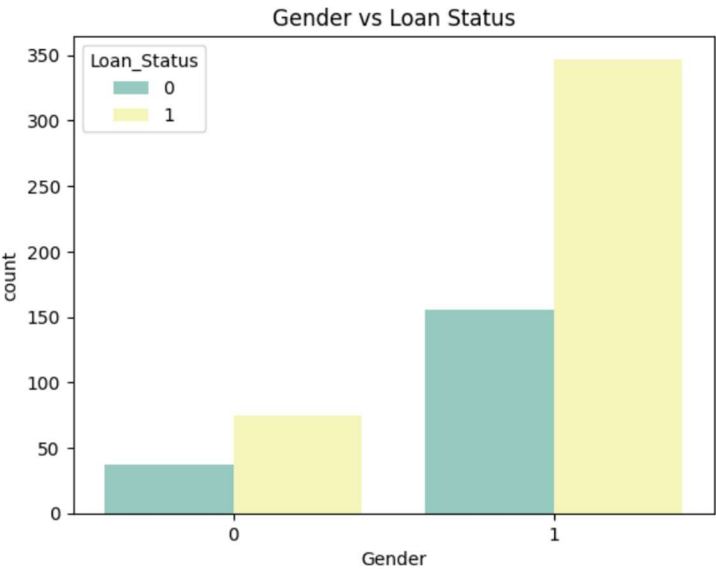


Figure 2: Gender-wise Loan Approval Comparison

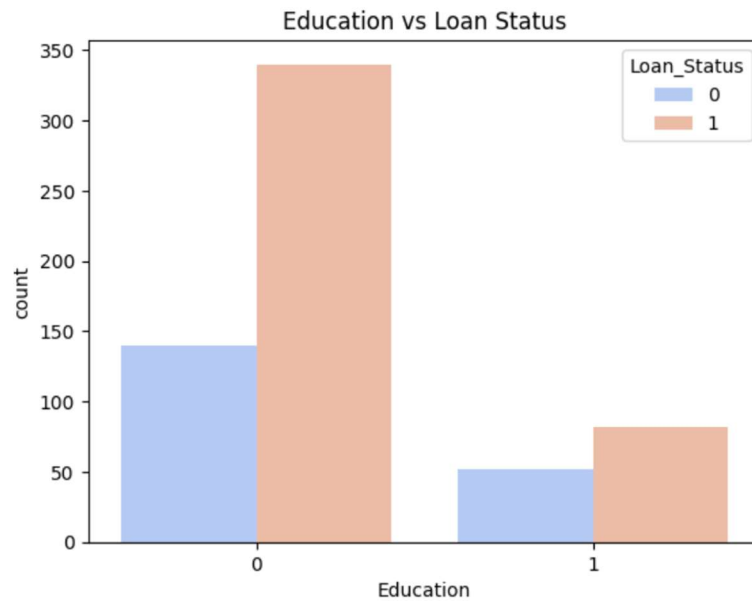


Figure 3: Education vs Loan Status Plot

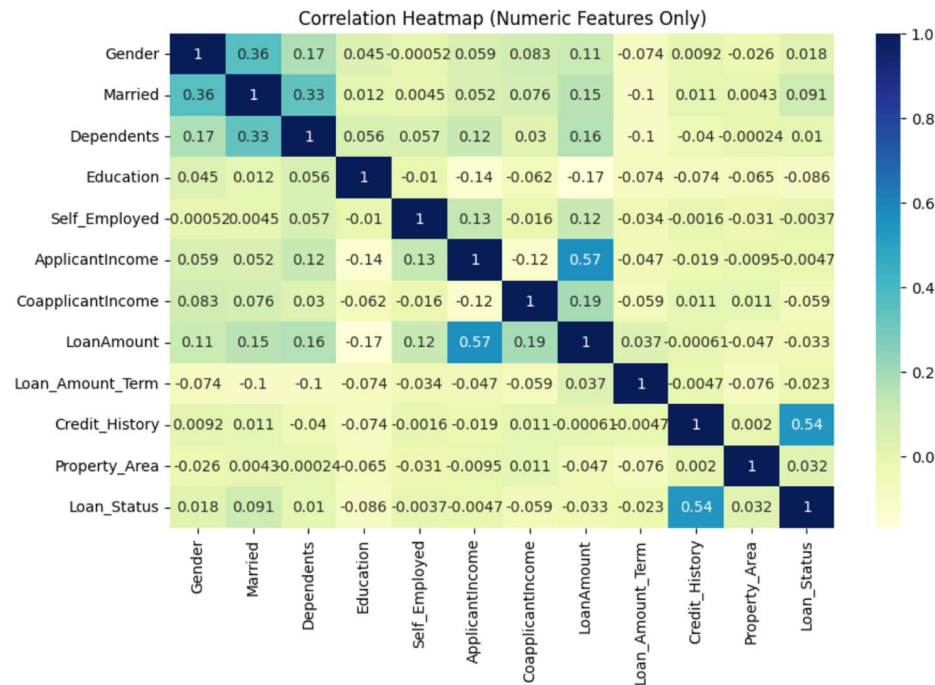


Figure 4: Correlation Heatmap Showing Relationships Among Variables



Figure 5: Confusion Matrix for Logistic Regression Model

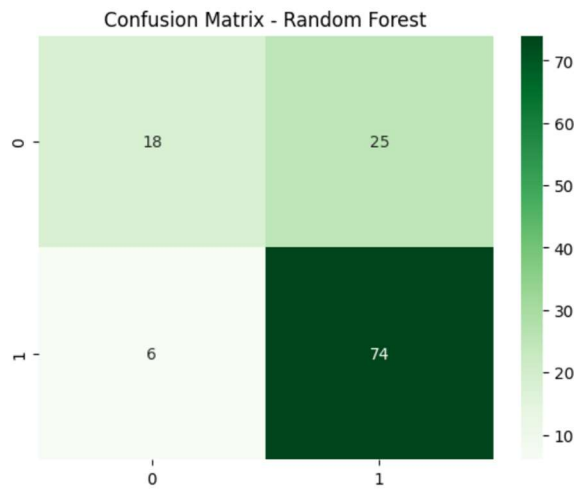


Figure 6: Confusion Matrix for Random Forest Model

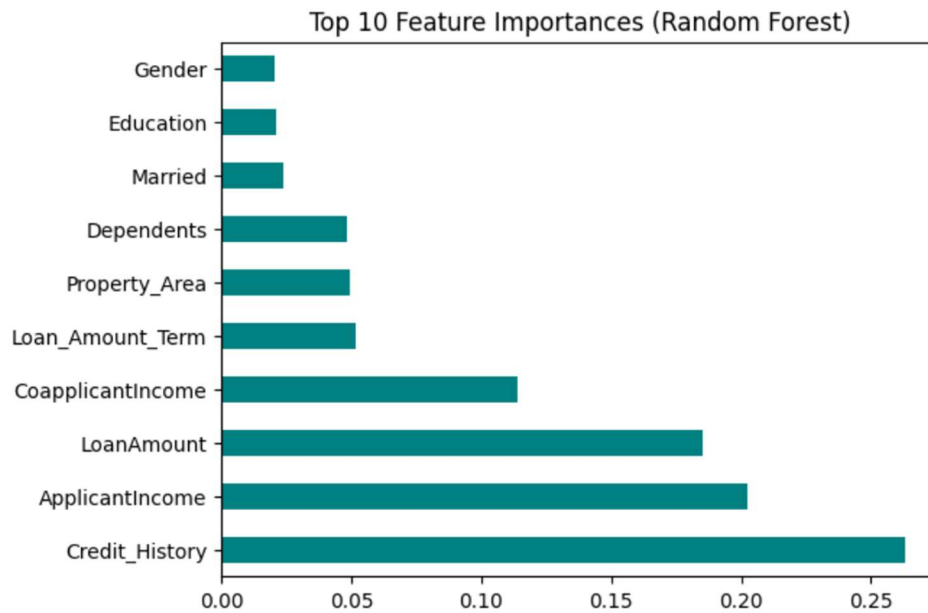


Figure 7: Feature Importance Graph from Random Forest Model

From the above visualizations, several key insights can be observed. The initial graphs, such as the *Loan Status Distribution*, *Gender-wise Approval*, and *Property Area vs Loan Status*, clearly indicate that applicants with a favorable Credit History are much more likely to have their loans approved. Male applicants were slightly higher in count, but gender was not a dominant factor influencing loan outcomes. Similarly, semi-urban applicants showed a higher approval rate compared to those in rural areas.

The Correlation Heatmap highlighted that Credit History, Applicant Income, and Loan Amount have the strongest relationships with the target variable (*Loan_Status*). Features like *Education* and *Self_Employed* showed weak or negligible correlations, implying that they have less influence on loan approval.

The Confusion Matrices for both Logistic Regression and Random Forest models revealed that Logistic Regression correctly classified more approved loans, while Random Forest performed better in identifying rejections after hyperparameter tuning. This demonstrates that ensemble methods can provide better generalization.

Finally, the Feature Importance Graph from the Random Forest model confirmed that *Credit History* is the most critical factor, followed by *Applicant Income* and *Loan Amount*. These features collectively contribute the most towards predicting whether a loan will be approved or not.

Overall, the visualization results align with logical financial reasoning — applicants with good credit records and stable incomes are more likely to receive loan approval.

Conclusion:

The *Loan Default Prediction System* successfully achieved its objective of predicting loan approval using data-driven methods. Through Logistic Regression and Random Forest algorithms, the project demonstrated the importance of feature selection, preprocessing, and model evaluation.

Key learnings from this project include:

- The strong influence of credit history and applicant income in determining loan eligibility.
- The role of data preprocessing in improving model performance.
- The value of visual analytics for interpreting model insights.

This project highlights the practical application of Machine Learning in the finance industry and how predictive analytics can help institutions minimize financial losses. Future scope includes deploying the trained model as a Streamlit-based web app or integrating it into an institution's internal decision system for real-time analysis.

References:

- Kaggle Dataset: <https://www.kaggle.com/datasets/ninzaami/loan-predication>
- Scikit-learn Documentation: <https://scikit-learn.org/>
- Seaborn Documentation: <https://seaborn.pydata.org/>
- Matplotlib Documentation: <https://matplotlib.org/>